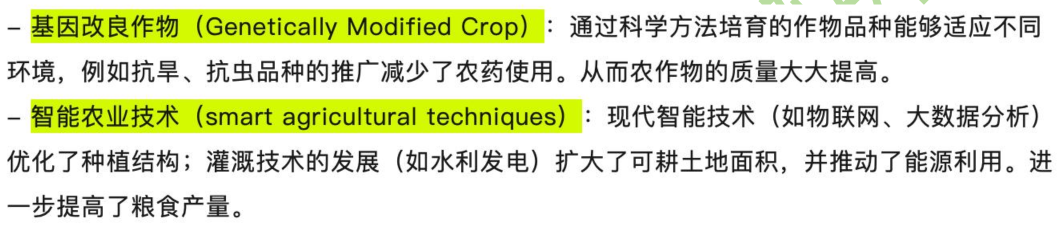
写作训练营 day 13

转基因

题目汇总：

With a growing world population, one of the most pressing issues is that of feeding such a large number of people. Some people think that GM foods offer a viable solution to this problem. To what extent do you agree or disagree?

Some people think it is better to use scientific techniques to grow food. Others, however, think that we should grow food naturally. Discuss both views and give your opinion.



- 基因改良作物（Genetically Modified Crop）：通过科学方法培育的作物品种能够适应不同环境，例如抗旱、抗虫品种的推广减少了农药使用。从而农作物的质量大大提高。
- 智能农业技术（smart agricultural techniques）：现代智能技术（如物联网、大数据分析）优化了种植结构；灌溉技术的发展（如水利发电）扩大了可耕土地面积，并推动了能源利用。进一步提高了粮食产量。

The range and quality of food has been improved with the development of technology and scientific advance. Some people think it is good and others think it is harmful. Discuss two sides and give your opinion.

转基因
读写互通海燕带你

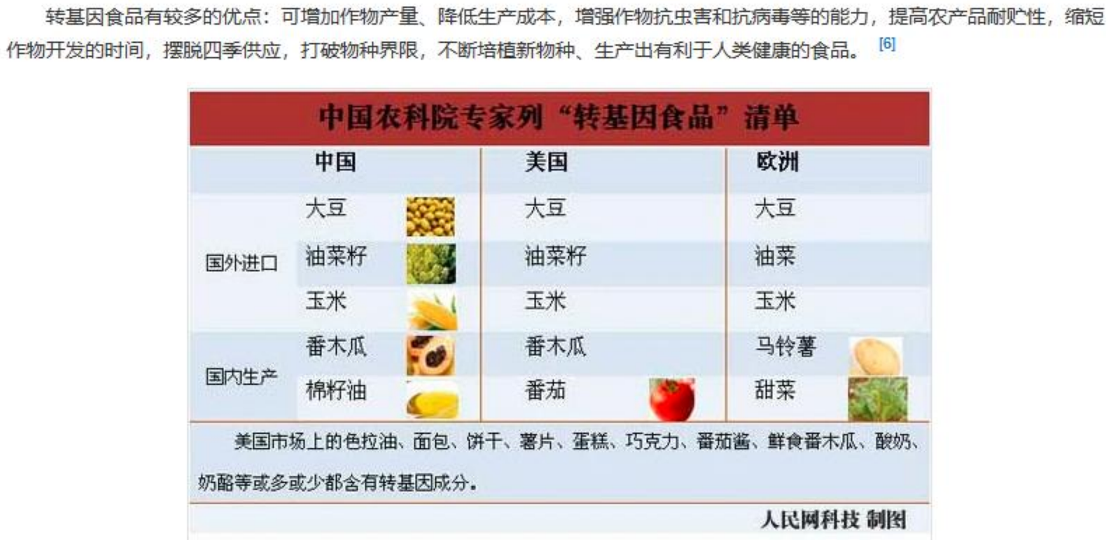
定义：

生物工程 Genetic engineering

Definition: Genetic engineering is the practice of manipulating the genes of an organism.

Specific meaning: Genetic engineering**,** the artificial manipulation, modification, and recombination of DNA or other nucleic acid molecules in order to modify an organism or population of organisms. The term *genetic engineering* is generally used to refer to methods of recombinant DNA technology, which emerged from basic research in microbial genetics. The techniques employed in genetic engineering have led to the production of medically important products, including human insulin, human growth hormone, and hepatitis 肝病 B vaccine, as well as to the development of genetically modified organisms such as disease-resistant plants.

转基因食品 GM -food



转基因食品有较多的优点：可增加作物产量、降低生产成本，增强作物抗虫害和抗病毒等的能力，提高农产品耐贮性，缩短作物开发的时间，摆脱四季供应，打破物种界限，不断培植新物种、生产出有利于人类健康的食品。[6]

中国农科院专家列“转基因食品”清单

	中国	美国	欧洲
国外进口	大豆	大豆	大豆
	油菜籽	油菜籽	油菜
	玉米	玉米	玉米
国内生产	番木瓜	番木瓜	马铃薯
	棉籽油	番茄	甜菜

美国市场上的色拉油、面包、饼干、薯片、蛋糕、巧克力、番茄酱、鲜食番木瓜、酸奶、奶酪等或多或少都含有转基因成分。

人民网科技 制图

常规思考角度:

优点

Increased Yield: Genetically modified crops often yield more due to enhanced resistance to pests and diseases, addressing food security concerns.

提高产量:由于转基因作物对病虫害的抵抗力增强，产量往往更高，从而解决了粮食安全问题。

Reduced Pesticide Use: Resistance to pests reduces reliance on pesticides, leading to environmental benefits by minimizing chemical impact.

减少农药的使用:对害虫的抵抗力减少了对农药的依赖，通过最大限度地减少化学影响来实现环境效益。

读写互通海燕带你

Improved Nutrition:Genetic modifications can boost crop nutritional content, helping combat deficiencies in populations reliant on these crops.

改善营养:基因改造可以提高作物的营养成分，帮助解决依赖这些作物的人群的营养不足问题。

Stress Resistance:Some crops are modified to **withstand environmental stresses** like **drought** and **salinity**, promoting sustainable agriculture in challenging conditions.

抗逆性:一些作物经过改良，可以承受干旱和盐度等环境压力，在具有挑战性的条件下促进可持续农业

Extended Shelf Life: Genetic modifications can enhance crop shelf life, reducing post- harvest losses and improving overall food availability.

延长保质期:基因改造可以延长作物的保质期，减少收获后的损失，提高总体粮食供应。

Customization: Genetic engineering allows **tailoring** crops for specific purposes like biofuel or pharmaceutical production.

定制:基因工程允许为特定目的定制作物，如生物燃料或制药生产。

缺点

读写互通海燕帮你搞定转基因

Environmental Risks: Widespread cultivation of genetically modified crops may pose environmental risks, such as harm to non-target organisms and the development of resistant pests.

环境风险:转基因作物的广泛种植可能带来环境风险，例如对非目标生物的危害和抗性害虫的发展。

Biodiversity Loss: Dominance of genetically modified varieties may lead to a reduction in biodiversity potentially making agriculture more vulnerable.

生物多样性丧失:转基因品种的优势地位可能导致生物多样性减少，可能使农业更加脆弱。

Unknown Long-Term Effects: The long-term impacts of genetically modified organisms on health and the environment are not fully understood.

未知的长期影响:转基因生物对健康和环境的长期影响尚未完全了解;

Corporate Control:Concerns exist about concentration of power in large biotech companies.limiting access for smaller farmers and restricting seed exchange.

企业控制:人们担心大型生物技术公司的权力集中，限制了小农户的准入，并限制了种子交换。

Ethical and Social issues: Genetically modified crops raise ethical questions and may impact cultural practices and traditions.

伦理和社会问题:转基因作物引发伦理问题，并可能影响文化习俗和传统。

读写互通海燕带你搞定转基因

补充材料

序号	中文	英文	考核
1	抵御不良环境	withstand adverse condition	
2	生存率更高	be likely to survive	
3	抗害虫	pest -resistant	
4	病毒感染	virus infestation	
5	终结...的问题	put an end to the ringspot problem	
6	耐逆性	stress resistance	
7	杀虫剂	pesticide	
8	节约杀虫剂的开支	save on pesticides	
9	经济好处	economic benefits	
10	长得快	grow faster	
11	产量高	increased \ greater yields	
12	更快的回报率	greater return on the farming investment	
13	基因多样	genetically diverse.	
14	基因太单一	genetically uniform	
15	移除	wipe out = eliminate	
16	环境上的挑战	environmental challenge	
17	隐藏成本	hidden cost	
18	向农民收更多的种子费用	charge tree farmers more for its seeds	
19	长得更狂野	grow more aggressively than natural trees do.	
20	物种入侵	species invasion	
21	挤走其他的树	crowding out the natural trees	

读写互通海燕带你搞定转基因

正方立场

Genetic modification, a process used to change an organism's genes and hence its characteristics, is now being used to improve trees. Through genetic modification, it is possible to create trees that produce more fruit, grow faster, or withstand adverse conditions. Planting genetically modified trees on a large scale promises to bring a number of benefits.

First, genetically modified trees are designed to be harder than natural trees; that is, they are more likely to survive than their unmodified counterparts. In Hawaii, for example, a new pest-resistant species of papaya tree has been developed in response to ringspot virus infestations that have repeatedly damaged the native papaya-tree population. Planting the genetically modified papayas has largely put an end to the ringspot problem.植物环斑病

Moreover, genetically modified trees promise to bring a number of economic benefits to those who grow them. Genetically modified trees tend to grow faster, give greater yields—of wood, fruit, or other products—and be hardier. This allows tree farmers to get faster and greater returns on their farming investment and save on pesticides as well.

Finally, the use of genetically modified trees can prevent overexploitation of wild trees. Because of the growing demand for firewood or building timber, many forests around the world are being cut down faster than they can be replaced. Introducing genetically modified trees—designed for fast growth and high yield in given geographic conditions—would satisfy the demand for wood in many of those areas and save the endangered native trees, which often include unique or rare species.

反方立场

Sure, there are some benefits to plant genetically modified trees, but, are these trees as really great as they first sound? When you examine the subject firstly, there are some serious problems and costs associated with genetically modified trees.

First, genetically modified trees may be resistant to one particular condition. But that doesn't necessarily ensure their survival. You see, a typical non modified trees' population is genetically diverse. That means that for most threatening conditions, or climate, insects and other pests, whatever, there will be at least some individual trees of any given species of tree that are resistant. So even if most of one kind of trees is killed, those few resistant trees will survive and ensure the survival of that species of tree. But genetically modified trees are genetically much more uniform. So if they're exposed to an environmental challenge they have not been designed for, they all die. So if the climate changes, the genetically modified trees will likely to be completely wiped out.

Now as to the second point, they're hidden costs associated with genetically modified trees. You see, the company that genetically modifies the tree can charge tree farmers more for its seeds than un-genetically modified trees would cost. Also, as you've grown the tree, you can't just collect the seeds and plant the new tree for free. By law, you have to pay the company every time you plant.

And finally, genetically modified trees might actually cause even more damage to the local wild trees. You see, genetically modified trees often grow more aggressively than natural trees do. And, genetically modified trees are typically planted among natural trees. As a result, the genetically modified trees out compete the native trees for resources, sunlight, soil, nutrients, and water, eventually crowding out the natural trees.

反方的观点 ＋ 理由 ＋ 展开 ＋ 怼回去
不支持转基因
有的人支持转基因----- economic benefits --- develop: 1. grow faster\ increased yields \ greater return on their farming investment 2. pest - resistant - no pesticide - save on pesticides + However, it is not the case. \ this is only the incomplete part of the picture \ it is a shallow view which fails to do sth. + elaborate: charge farmers more : seed --- cannot collect seeds --- by law: every time you plant, you should pay for the seed

文章 2:
正方立场

Broccoli is a vegetable that is popular throughout the United States, but it can be grown only in temperate climates with mild summers. Because of this limitation, 90 percent of the broccoli consumed in the United States is grown in the cooler regions of California, on the West Coast. It must be shipped 4,000 kilometers across the entire country to reach the populous East Coast. This situation may soon change, however: scientists are attempting to create a new type of broccoli that can be grown on the East Coast despite the hotter summers there. This project has several potential benefits.

First, growing broccoli on the East Coast and distributing it to nearby grocery stores would lower broccoli costs for consumers. Transporting the vegetable from California by truck takes several days, which makes transportation costs high. Growing broccoli locally would reduce transportation costs, so stores could sell broccoli for less money.

Second, the new type of broccoli will probably be much more desirable to consumers than some other new crops. The new broccoli is being created using traditional breeding techniques: researchers are crossbreeding broccoli with other similar plants that have desirable characteristics, such as resistance to hot summers. Other new crops—varieties of squash or maize, for example—were created using genetic modification technology. Genetically modified crops are very controversial: many consumers believe they are unsafe and so reject them. However, the new broccoli will not be in that category.

Third, vegetables are healthiest to eat right after they are harvested because they retain most of their beneficial nutrients and vitamins. The longer it takes for a vegetable like broccoli to reach consumers, the fewer nutrients and vitamins it contains. Broccoli that is grown locally can be eaten much sooner after it is cut, which will make this new broccoli healthier for East Coast residents than California broccoli.

读写互通海燕帮你搞定转基因

远距离运输食品有坏处 -- nutrients and vitamins the vegetable contains -- loss after long distance transportation --- 对比论证： after they are harvested -- vegetables are healthiest -- retain most of their beneficial nutrients and vitamins

反方立场

East coast broccoli might not actually be as successful or beneficial as its creators claim.

First, it's true that growing and selling crops locally saves money on transportation costs, but that will not necessarily make the new broccoli cheaper for consumers. Yes, the new breed of broccoli will be able to survive the hot summers on the east coast, but unfortunately it will also have lower yields than the broccoli grown on the west coast. This means that farmers on the east coast will harvest less broccoli per hectare, and so they'll have to sell their produce at a higher price. This price increase will cancel out the savings on transportation, so the east coast consumers will probably not save any money buying the locally grown broccoli.

Second, even though researchers are using traditional breeding techniques in creating the new broccoli, the public might still be suspicious. You see, financial support for this research comes from the same companies that created genetically modified crops in the past. Because those companies used genetic modification techniques with other crops, many American consumers distrust those companies and might distrust any project that those companies are involved in. This might lead the consumers to reject the new broccoli as well.

Third, it's true that fresh broccoli has the greatest amount of health-promoting nutrients and vitamins, but is that a good argument for creating a new breed of broccoli that can be grown on the east coast? Not necessarily. Other vegetables that are already grown on the east coast contain similar nutrients that broccoli contains. For example, kale, a leafy green vegetable, provides comparable health benefits. And kale is in some respects better. It's grown on the east coast not only during the summer, but also during the fall and winter. Instead of wasting money and resources on creating new vegetables, we should educate people about eating fresh healthy vegetables that are already available locally.

作业 1: 上传听课笔记

作业 2: 整理逻辑链条 --- 转基因食品的利弊 长距离运输食品的利弊

作业 3: 多读外刊--相关主题发送到营地 --- 1 人找一篇， 针对额外的补充材料进行逻辑链条的补充

作业 4: 完成一整篇文章的创作 --- 3 选 1 -- 具体的题目在开头